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Door for mounting a removable electronic display

Abstract

In one general aspect, the subject matter described in this specification can be embodied in a display case door that includes an insulated panel assembly, a door frame, and an electronic display mounted to the door. The door frame extends about and is secured to a peripheral edge of the insulated panel assembly. The electronic display overlays the insulated panel assembly. A mounting bezel is secured to the door frame and has a coupling end that extends over an edge of the electronic display securing the electronic display to the door frame. A cover bezel is coupled to a front surface of the door frame, where the cover bezel is sized to overlay the front surface of the door frame and the coupling end of the mounting bezel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. application Ser. No. 17/890,126, filed on Aug. 17, 2022, which is a continuation of and claims priority to U.S. application Ser. No. 17/064,346, filed on Oct. 6, 2020, now U.S. Pat. No. 11,435,777, which is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. application Ser. No. 16/712,192, filed on Dec. 12, 2019, now U.S. Pat. No. 10,838,453, which in turn is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. application Ser. No. 16/369,054, filed on Mar. 29, 2019, now U.S. Pat. No. 10,514,722, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

(1) This invention relates to thermally insulated doors for temperature-controlled enclosures.

BACKGROUND

(2) Refrigerated enclosures are used in commercial, institutional, and residential applications for storing and/or displaying refrigerated or frozen objects. Refrigerated enclosures may be maintained at temperatures above freezing (e.g., a refrigerator) or at temperatures below freezing (e.g., a freezer). Refrigerated enclosures have one or more thermally insulated doors or windows for viewing and accessing refrigerated or frozen objects within a temperature-controlled space. Doors for refrigerated enclosures generally include thermally insulated glass panel assemblies.

(3) Displaying content on refrigerated display case doors can be an effective way of advertising products. Refrigerated display case doors with embedded display panels can be costly and difficult to maintain. Improvement in the methods for displaying content in refrigerated display case doors are continuously sought.

SUMMARY

(4) In one general aspect, the subject matter described in this specification can be embodied in a display case door that includes an insulated panel assembly, a door frame, and, an electronic display. The door frame extends about and is secured to a peripheral edge of the insulated panel assembly. The electronic display overlays the insulated panel assembly. A mounting bezel is secured to the door frame and has a coupling end that extends over an edge of the electronic display securing the electronic display to the door frame. A transparent protective panel is positioned between a front surface of the electronic display and the coupling end of the mounting bezel. A first cover bezel is coupled to a front surface of the door frame, where the first cover bezel is sized to overlay the front surface of the door frame and the coupling end of the mounting bezel. A circuitry module is positioned in a bottom portion of the door frame and overlays a portion of the insulated panel assembly. The circuitry module is releasably coupled to the door frame. The circuitry module includes a media player in electronic communication with the electronic display, where the media player is configured to control media content presented on the electronic display. A power cable is disposed within a channel defined by a portion of the door frame and the power cable is electrically connected to the electronic display and to the media player. A second bezel cover is coupled to the door frame and covers the circuitry module. The second bezel cover includes a plurality of louvers configured to provide cooling air-flow to the media player.

(5) This and other implementations can each optionally include one or more of the following features.

(6) In some implementations, the insulated panel assembly includes an opaque insulating material. In some implementations, the insulated panel assembly includes foam.

(7) In some implementations, the insulated panel assembly is a foam filled panel.

(8) In some implementations, the electronic display overlays the insulated panel assembly and

overlaps more than 50% of a surface area of the insulated panel assembly.

(9) In some implementations, the electronic display and the circuitry module together overlay the insulated panel assembly and overlap an entire surface area of the insulated panel assembly.

(10) In some implementations, the mounting bezel includes two parallel, vertical members. In some implementations, the coupling end of the mounting bezel includes a flange of each vertical member and a flange of at least one horizontal member, each respective flange extending generally parallel to the insulated panel assembly. In some implementations, a coupling end of each vertical member and a coupling end of one horizontal member extend over the edge of the electronic display overlapping three edges of a housing of the electronic display. In some implementations, each of the vertical members include at least one mounting bracket configured to secure the mounting bezel to the door frame.

(11) In another general aspect, the subject matter described in this specification can be embodied in a display case door that includes an insulated panel assembly, a door frame, and an electronic display mounted to the door. The door frame extends about and is secured to a peripheral edge of the insulated panel assembly. The electronic display overlays the insulated panel assembly. A mounting bezel is secured to the door frame and has a coupling end that extends over an edge of the electronic display securing the electronic display to the door frame. A cover bezel is coupled to a front surface of the door frame, where the cover bezel is sized to overlay the front surface of the door frame and the coupling end of the mounting bezel.

(12) In another general aspect, the subject matter described in this specification can be embodied in a display case door that includes an insulated panel assembly and a door frame extending about and secured to a peripheral edge of the insulated panel assembly. The door frame is configured to support an electronic display mounted thereto. The door frame includes a portion that extends outwards a distance beyond a front surface of the insulated panel assembly, and a plurality of mounting sites arranged along the portion of the door frame. The mounting sites are configured to interface with a display mounting retainer. The electronic display, when mounted, overlays the insulated panel assembly. The distance that the portion extends beyond the front surface of the insulated panel assembly corresponds to a thickness of the electronic display such that, when mounted, a front surface of the electronic display is substantially flush with a front surface of the door frame.

(13) These and other implementations can each optionally include one or more of the following features.

(14) In some implementations, the insulated panel assembly includes an opaque insulating material. In some implementations, the insulated panel assembly includes foam. In some implementations, the insulated panel assembly is a foam filled panel.

(15) In some implementations, the electronic display, when mounted, overlays a majority of the insulated panel assembly.

(16) Some implementations include a transparent protective panel positioned between a front surface of the electronic display and the coupling end of the mounting bezel.

(17) Some implementations include a circuitry module positioned in a bottom portion of the door frame and overlaying a portion of the insulated panel assembly, where the circuitry module is releasably coupled to the door frame, and the circuitry module includes a media player in electronic communication with the electronic display. The media player is configured to control media content presented on the electronic display.

(18) Some implementations include a second cover bezel coupled to the door frame and covering the circuitry module. In some implementations, the second cover bezel includes a plurality of louvers configured to provide cooling air-flow to the media player.

(19) Particular implementations of the subject matter described in this specification can be implemented so as to realize one or more of the following advantages.

(20) Implementations may provide less expensive alternatives to doors with embedded displays.

Implementations may improve the maintainability of electronic displays mounted to doors. For example, implementations of the present disclosure may provide for easy replacement or maintenance of electronic displays mounted to a display case door without the need to remove an entire display case door. Implementations of the present disclosure may also allow a display case door with mounted electronic displays to stay close to the original envelope (e.g., the door having a thin profile) so the door does not interfere with the motion of other doors in a continuous lineup of doors. For example, implementations may permit display case doors having electronic displays to be mounted in a continuous lineup of doors (e.g., in a supermarket aisle) without the need to alter the mounting configuration or spacing between doors to accommodate the thickness of the electronic displays.

(21) The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a front view of an exemplary arrangement of electronic displays in a display case door according to implementations of the present disclosure.

(2) FIG. 2 is a front view of an exemplary arrangement of electronic displays in a display case door according to implementations of the present disclosure.

(3) FIG. 3 is a front view of an exemplary arrangement of electronic displays in a display case door according to implementations of the present disclosure.

(4) FIG. 4 is a front view of two exemplary display case doors mounted to a display case in a first arrangement.

(5) FIG. 5 is a front view of two exemplary display case doors mounted to a display case in a second arrangement.

(6) FIG. 6 is a cross-sectional view taken along line 6-6 of a portion of the display case door shown in FIG. 4 according to a first exemplary embodiment.

(7) FIG. 7 is a cross-sectional view of the portion of the display case door shown in FIG. 6 without electronic displays.

(8) FIG. 8A is a cross-sectional view of the portion of the display case door shown in FIG. 6 according to a second exemplary embodiment.

(9) FIG. 8B is a cross-sectional view of the portion of the display case door shown in FIG. 8A, according to a third exemplary embodiment.

(10) FIG. 8C is a cross-sectional view of the portion of the display case door shown in FIG. 8A, according to a fourth exemplary embodiment.

(11) FIG. 9 is a cross-sectional view of the portion of the display case door shown in FIG. 6, according to a fifth exemplary embodiment.

(12) FIG. 10 is a cross-sectional view of a portion of the display case door shown in FIG. 4, taken along line 10-10 in FIG. 4.

(13) FIG. 11 is a cross-sectional view of the portion of the display case door shown in FIG. 10, according to a second exemplary embodiment.

(14) FIG. 12 is a cross-sectional view of the portion of the display case door shown in FIG. 8A, according to a sixth exemplary embodiment.

(15) FIG. 13 is a schematic top view of two adjacent doors, showing the motion of a door.

(16) FIG. 14 is a front view of another exemplary display case door according to implementations of the present disclosure.

(17) FIG. 15 is a front view of the display case door in FIG. 14, showing mounting arms and a

media player according to a first exemplary embodiment.

(18) FIG. **16** is a front view of a display case door showing a media player according to a second exemplary embodiment.

(19) FIG. **17** is a front view of an exemplary arrangement of electronic displays in a display case door according to implementations of the present disclosure.

(20) FIG. **18** is a cross-sectional view taken along line **18-18** of a portion of the display case door shown in FIG. **14**, according to a first exemplary embodiment.

(21) FIG. **19** is a cross-sectional view of the portion of the display case door shown in FIG. **18**, according to a second exemplary embodiment.

(22) FIG. **20** is an electrical diagram of a display case door assembly according to implementations of the present disclosure.

(23) FIG. **21** is an electrical diagram of a display case door assembly according to implementations of the present disclosure.

(24) FIG. **22** is a front view of another exemplary display case door according to implementations of the present disclosure.

(25) FIG. **23A** is a front view of an exemplary bezel cover according to implementations of the present disclosure.

(26) FIG. **23B** is a cross-section view of the bezel cover shown in FIG. **23A**.

(27) FIG. **24** is a front view of another exemplary display case door according to implementations of the present disclosure.

(28) FIG. **25** is an exploded view of the display case door shown in FIG. **24**.

(29) FIG. **26A** is a front view of another exemplary display case door according to implementations of the present disclosure.

(30) FIG. **26B** is a cross-sectional view of a portion of the display case door shown in FIG. **26A**, taken along line **26B-26B** in FIG. **26A**.

(31) FIG. **27** is a cross-sectional view of a portion of the display case door shown in FIG. **24**, taken along line **27-27** in FIG. **24**.

(32) FIG. **28** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a second exemplary embodiment.

(33) FIG. **29** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a third exemplary embodiment.

(34) FIG. **30** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a fourth exemplary embodiment.

(35) FIG. **31** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a fifth exemplary embodiment.

(36) FIG. **32** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a sixth exemplary embodiment.

(37) FIG. **33** is a cross-sectional view of the portion of the display case door shown in FIG. **27**, according to a seventh exemplary embodiment.

(38) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(39) FIG. **1** is a front view of an exemplary arrangement of electronic displays **18a** in a display case door **10** according to implementations of the present disclosure. FIG. **1** illustrates an exemplary display case door **10** configured to be installed in a refrigerated display case such as a refrigerator, a freezer, or other enclosure defining a temperature-controlled space. Display case door **10** includes an insulated panel assembly **11** (shown in FIG. **6**), a door frame **12**, display retainers **14**, and electronic displays **18a** overlaying panel assembly **11**.

(40) As further discussed in more detail below with respect to FIGS. **24** and **25**, the display case door can include an insulated panel assembly, a door frame secured to an edge of the panel assembly, and an opaque display screen (e.g., an electronic display) coupled to the door frame. In

some examples, the insulated panel assembly can be an opaque insulating material such as a foam filled panel. In other examples, the insulated panel assembly can be a transparent insulated glass package. The display screen can be mounted to the door frame so as to overlay a majority of the insulated panel assembly. The electronic display can be secured to the door frame by at least one display retainer secured to the door frame. For example, the display retainer has a coupling end that mates with the electronic display to releasably secure the electronic display to the door frame. Additionally, as further discussed in detail below with respect to FIG. 26, a display case door can be made without the electronic display included. For example, the display case door can include an insulated panel assembly and a door frame that is configured to support an electronic display mounted on the door. For example, the door frame of the display case door can include a portion that extends outwards a distance beyond a front surface of the insulated panel assembly. The door frame can also have multiple mounting sites arranged along the portion of the door frame that are configured to interface with a display retainer. The door frame can be configured such that when mounted, the electronic display overlays a majority of the insulated panel assembly. The distance that the portion of the door frame extends beyond the front surface of the panel assembly can correspond to a thickness of the electronic display such that, when mounted, a front surface of the electronic display is substantially flush with a front surface of the door frame.

(41) As discussed in more detail below, panel assembly **11** includes one or more panes of glass. In some implementations, panel assembly **11** includes an opaque thermally insulating panel (e.g., a foam filled panel). In other implementations, panel assembly **11** includes two or more layers of transparent panes bounding a sealed space in between, forming a sealed glass unit (SGU). Door frame **12** extends around and is secured to a peripheral edge of panel assembly **11**. Frame **12** defines a channel or tunnel (not shown) configured to receive one or more power cables that provide electrical power to the electronic displays **18a** (e.g., monitors).

(42) Display retainers **14** releasably secure electronic displays **18a** to the frame **12** of the display case door **10**. Display retainers **14** can be made of an extruded metal or hard plastic, such as aluminum or polyvinyl chloride (PVC). Display retainers **14** can be made in different sizes and configuration to mount electronic displays **18a** of different thicknesses. Referring briefly to FIG. 6, each display retainer **14** has a first end **19** and a second end **21** (e.g., a coupling end). The first end **19** is arranged to couple to or clip to the frame **12**. The second end **21** (e.g., coupling end) configured to extend over an edge **20** of each electronic display **18** to releasably secure electronic displays **18** to frame **12**. Referring back to FIG. 1, in some implementations door **10** can include four display retainers **14**, each having a length similar to a respective side of frame **12**. In some embodiments, display retainers **14** can be shorter than their respective side of frame **12**, allowing multiple display retainers **14** along a side of frame **12** to mount electronic displays **18a** to the door **10**.

(43) Electronic displays **18a** include an outer housing **37** and an electronically controllable display panel **39** (e.g., screen) mounted to outer housing **37**. Outer housing **37** can be positioned directly against the front surface of panel **11**. In some examples, a protecting foam or flexible tape can separate outer housing **37** from the front surface of panel **11**. Electronic displays **18a** can include, but are not limited to, liquid crystal displays (LCD), light emitting diode (LED) displays, organic light emitting diode (OLED) displays, field emission displays (FED), plasma display panels (PDP), or electroluminescent (EL) displays. For example, electronic displays **18a** can be smart televisions with streaming capabilities for receiving content over a wireless network (e.g., a Wi-Fi network).

(44) Display retainers **14** allow electronic displays to be quickly replaced or removed for maintenance or replacement. Retainers **14** also allow electronic displays to be re-arranged in different configurations. Retainers **14** allow electronic displays **18a** to be mounted to door **10** without need for special mounting equipment or reconfiguration of the electronic displays housing **37**. In the example arrangement shown in FIG. 1, electronic displays **18a** are mounted in a one by four arrangement. In other words, the four electronic displays **18a** are mounted to door **10** in one

column with four rows of displays. As further discussed below, electronic displays **18a** can be mounted in different arrangements. For example, multiple electronic displays can be mounted over panel assembly **11**, with electronic displays arranged in one column that extends along a height of the panel assembly. In some implementations, multiple electronic displays can be mounted over the panel assembly, with the electronic displays arranged in two or more columns that extend along a height of the panel assembly.

(45) Frame **12** can include four frame rails **13** each made of extruded aluminum, PVC, or a similar material. Frame rails **13** can each have a constant cross-section along their length. One of the frame rails, e.g., frame rail **13a**, also referred to herein as hinge rail, can define the channel through which the power cord extends to power electronic displays **18a**. As discussed in more detail below, frame **12** further defines openings at a top and bottom thereof for hingedly connecting door **10** to a display case frame. Such openings can be defined at the top and bottom ends of the frame rail channel. The hinges used to connect door can include an electrical connector from which the power cable extends.

(46) Door frame **12** has a width and thickness that allows display case door **10** to be installed and operated in an existing refrigerated display case without the need of retrofitting the display case. Door **10** can be operated with or without electronic displays **18a**. In some examples, electronic displays **18a**, when installed, overlap 75% or more of a surface area of the insulated panel assembly **11**.

(47) In some implementations, panel assembly **11** is an opaque thermally insulating panel. For example, panel assembly **11** can include a foam filled insulating panel.

(48) In some implementations, panel assembly **11** includes one or more panes of transparent or substantially transparent glass (e.g., insulated glass, non-tempered glass, tempered glass, etc.), plastics, or other transparent or substantially transparent materials (e.g., translucent materials). The panel assembly can be a glass or plastic panel that is transparent to visible light. The transparent panel assembly can be a highly insulating panel, e.g., a panel having an R value of 2 or more, and a U value of less than 0.5. As further discussed in detail with respect to FIG. 6A, panel assembly **11** can include multiple layers of transparent panes. For example, panel assembly **11** can be a multi-pane unit having a first pane and a second pane that are separated by a gap, forming an SGU. Panel **11** can be a thin SGU assembly having a thickness of 1-3 centimeters. Panel **11** can have an insulation R-value of between 2 and 30 or, in some examples, between 4 and 8. The SGU can be a highly insulating panel assembly, e.g., an assembly having an R value of 2 or more, and a U value of less than 0.5. It is not necessary that panel **11** be transparent. In some implementations, panel **11** can be translucent or opaque.

(49) In some implementations, the gap or sealed space can between two or more panels can be filled with an insulating gas such as a noble gas (e.g., Argon, Krypton, etc.) which functions as a thermal insulator to reduce heat transfer through the panel. In some examples, the sealed space can be evacuated below atmospheric pressure. For example, panel assembly **11** can be a vacuum insulated glass (VIG) assembly. For example, a VIG assembly is a SGU in which the sealed space can be an evacuated space below atmospheric pressure. As further discussed in FIG. 8A, panel assembly **11** can be an offset glass package panel assembly. For example, a pane contained within frame **12** can include multiple smaller panes attached to a back surface of the secured pane.

(50) Display case door **10** includes a door handle **16** coupled to frame **12**. Door handle **16** can be made from extruded aluminum tubes that are cut to a specified dimension and coupled to a front or side surface of frame **12**. Door handle **16** can be a 'pull handle' of different shapes. For example, handle **16** can be a U-shaped handle or a handle with a J-shape cross-section. Handle **16** can be attached to frame **12** by using an adhesive or a mechanical fastener.

(51) Referring to FIGS. 1-3, electronic displays **18a**, **18b**, and **18c** can be mounted to door **10** in various different arrangements. For example, FIG. 1 shows a display case door **10** according to a first embodiment, having electronic displays **18a** mounted in a one by four configuration. FIG. 2

shows another example arrangement of electronic displays mounted on door **10**. FIG. 2 shows a display case door **10** according to a second embodiment, having electronic displays **18b** mounted in a one by two configuration. Similar to how the electronic displays **18a** in FIG. 1 are mounted on door **10**, electronic display retainers **14** extend over an edge **20b** of electronic displays **18b** to releasably secure electronic displays **18b** to the frame **12** of door **10**.

(52) FIG. 3 shows another example arrangement of electronic displays mounted on door **10**. FIG. 3 shows a display case door **10** according to a third embodiment, having electronic displays **18c** mounted in a two by two configuration. In other words, FIG. 3 shows four electronic displays **18c** mounted to door **10** in two columns and two rows of displays. Similarly, display retainers **14** extend over an edge **20c** of each electronic display **18c** to releasably secure electronic displays **18c** to the frame **12** of door **10**.

(53) Implementations of the present disclosure provide flexibility in the arrangement of electronic displays on a display case door. For example, the display retainers described herein permit various arrangements of electronic displays. For example, multiple electronic displays can be mounted over panel assembly **11**, with electronic displays arranged in one column that extends along a height of the panel assembly **11**. In some implementations, multiple electronic displays can be mounted over the panel assembly **11**, with the electronic displays arranged in two columns that extend along a height of the panel assembly **11**.

(54) In some implementations, all portions of electronic displays **18a**, **18b**, and **18c** can overlay panel assembly **11**. For example, the electronic displays can be arranged so as to overlap a majority of the surface area of panel assembly **11**. In some implementations, electronic displays **18a** are arranged to overlap the entire surface area of panel assembly **11**. In some examples, electronic displays **18a** can cover the front surface of panel assembly **11**, leaving a small gap **22** between adjacent electronic displays **18a**. In some examples, a flexible foam or tape (not shown) can fill the gap **22**, acting as a cushion between adjacent electronic displays. In some embodiments, electronic displays **18a** can be arranged to expose one or more areas of the front surface of panel assembly **11**. In some implementations, display retainers can be arranged so as to form a venting gap between adjacent display retainers, for air to enter to and ventilate the back of electronic displays **18a**.

(55) FIG. 4 is a front view of two exemplary display case doors **10** mounted to a display case **100** in a first arrangement. Refrigerated display case **100** can be a refrigerator, freezer, or other enclosure defining a temperature-controlled space. For example, refrigerated display case **100** can be a refrigerated display case or refrigerated merchandiser in grocery stores, supermarkets, convenience stores, florist shops, and/or other commercial settings to store and display temperature-sensitive consumer goods (e.g., food products and the like). Refrigerated display case **100** can be used to display products that must be stored at relatively low temperatures. In some implementations, refrigerated display case **100** is a refrigerated display unit used, for example, in warehouses, restaurants, and lounges. For example, refrigerated display case **100** can be a free-standing unit or “built-in” unit that forms a part of the building in which the refrigerated display case **100** is located.

(56) As illustrated, refrigerated display case **100** has multiple display case doors **10** mounted on a display case frame **17**. Each door **10** is pivotally mounted on hinges **29** that connect the door **10** to frame **17**. Hinges **29** of each door **10** are in opposite ends of case frame **17**, allowing doors **10** to pivot about points located at two opposite ends of case frame **17**. In such configuration, handles **16'** are adjacent to one another. In some implementations, each handle **16'** is attached to a side surface of a respective door frame **12**. For example, handles **16'** can have a J-shaped cross-section, in which a flange of handle **16'** is attached to a side surface of frame **12**. In some examples, the handles can be U-shaped handles attached to a front surface of frame **12**. In some implementations, doors **10** can be sliding doors configured to open and close by sliding with respect to case frame **17**. For example, hinges **29** can be replaced by a pair of corresponding rails coupled, respectively, to each door **10** and case frame **17**.

(57) FIG. 5 is a front view of two exemplary display case doors **10** mounted to a display case **100** in a second arrangement. For example, doors **10** are installed in an arrangement in which hinges **29** are placed on the same side of each door **10**. Doors **10** open in the same direction, which may be advantageous for a long row of doors (e.g., in a supermarket aisle). For example, the arrangement shown in FIG. 5 may leave more room for consumers to open doors **10**, as opposed to doors with adjacent handles such as in the example shown in FIG. 4.

(58) FIG. 6 is a cross-sectional view taken along line 6-6 of a portion of the display case door **10** shown in FIG. 4. Frame **12** includes a frame rail **13a** and a display retainer **14**. Frame rail **13a** defines a channel **33** that extends along the length of the frame rail **13a**. Channel **33** can have a channel opening along the length of frame rail **13a** that allows power cables **50** to extend away from channel **33** to electronic displays **18**.

(59) In some implementations, frame rail **13a** can serve as a hinge rail. For example, a bottom opening **35** of frame rail **13a** can be configured to receive a bottom hinge **29a** (e.g., a pin hinge). Hinge **29a** can be attached to the frame of case **100** and inserted into the bottom of opening **35**. Bottom hinge **29a** can be an electrical hinge from which one or more power cables **50** extend. For example, hinge **29a** can include an electrical connector **54** from which each power cable **50** extends. In some examples, electrical connector **54** includes a 120V electrical plug. Electrical plug **54** can include a sleeve **55** that extends into channel **33** to cover power cable **50**, and is configured to guide cable **50** into channel **33**. In some implementations, bottom hinge **29a** can be a gravity hinge and the top hinge (not shown) can be an electrical hinge from which power cables **50** extend. In some implementations, other electrical cables can extend from hinge **29a**. For example, panel assembly **11** may include an anti-condensation system that requires power for heating the panel **11** (e.g., providing power to an electro-conductive coating), in which additional electrical conductors may provide such power. In some implementations, instead of hinges **29**, the display case door can be mounted to case **100** using any type of appropriate hinges such as knuckle hinges or ball bearing hinges.

(60) In the example arrangement shown in FIG. 6, display retainer **14** is an over-center clip with an S-shaped cross-section. Display retainer **14** can be a self-locking longitudinal clip that zips, for example, to electronic display edge **20** from a top of electronic display **18** to a bottom, or from bottom to top. In some implementations, display retainer **14** can be a hard plastic that is slightly bendable to clip over electronic display frame **20**. Display retainer **14** includes a first end **19** configured to clip to inner surface **56** of frame rail **13a**. Display retainer **14** includes a coupling end **21** opposite first end **19**, extending offset from frame **12**. Coupling end **21** is spaced away from a peripheral frame edge **58** of door frame **12**, and away from front surface **60** of panel **11**. Coupling end **21** is configured to extend over electronic display edge **20** to secure electronic display **18** to frame **12**. In some implementations, display retainer **14** can clamp or wrap the electronic display edge **20** as display retainer **14** is rotated into position. For example, as display retainer rotates about edge **58** from a generally diagonal position (i.e., disengaged position) to a vertical position, coupling end **21** begins to engage electronic display edge **20**. Display retainer **14** provides a clamping force between frame rail **13a** and the edge **20** of electronic display. Display retainer can be secured to the inner surface **56** of frame rail **13a** by the clamping force. In some implementations, the first end **19** of display retainer **14** can be attached to inner surface **56** with an adhesive.

(61) To remove an electronic display **18**, display retainer **14** can be rotated back from a vertical position to a diagonal position. A bottom display retainer **14a** can secure a bottom edge of a bottom electronic display in a similar way. Similarly, display retainers **14** along the rest of door frame **12** can secure each electronic display **18** into place.

(62) Electronic displays **18** can be positioned directly against front surface **60** of panel **11**. For example, a back surface of outer housing **37** can be sufficiently soft to prevent or reduce scratching of panel **11** when electronic displays **18** come into contact with panel **11**. In some implementation,

a protecting foam or flexible tape **31** can separate outer housing **37** from front surface **60** of panel **11**.

(63) Frame rail **13a** further includes a channel **52** into which panel **11** is secured to frame **12**. Panel **11** can be bonded to frame **12** within channel **52** with an adhesive, such as an epoxy or polyurethane. A spacer **32** or an adhesive such as acrylic can also be used to secure panel **11** to frame **12**. In some examples, a mechanical fastener such as a clamp may be used to secure panel **11** to frame **12**. Display case **100** includes other structures for attaching door to display case **100**, such as mullions, gaskets **34**, and other associated hardware often included in display case doors.

(64) FIG. **7** is a cross-sectional view of the portion of the display case door **10** shown in FIG. **6**. FIG. **7** illustrates a door **10** without electronic displays mounted to it. The electronic displays can be removed from door **10** without impairing the functionality of door **10**. The electronic displays can be quickly removed by disengaging the retainers **14** from the electronic display **18**. For example, the retainers **14** can be disengaged by unzipping, unclipping, or removing the over-center clips from the electronic display edges. Frame **12** can receive a frame plug **15** that covers power cables **50** and electrical connector **54**. Frame plug **15** can be a flexible plastic strip with a constant cross-section configured to snug into the opening formed between frame edge **58** and the panel receiving portion **52** of frame rail **13a**. Plug **15** may protect cable **50** and connector **54** from the outside environment and may improve the aesthetic appearance of frame **12**.

(65) FIG. **8A** depicts a detail view of the portion of the display case door shown in FIG. **6**, according to another example implementation. Similar to the frame rail in FIG. **6**, frame rail **113** is configured to receive bottom hinge **29a** and power cable **50** within channel **133**. In the example arrangement shown in FIG. **8A**, display retainer **114** is a door trim made of hard plastic (e.g., PVC) or an extruded metal such as aluminum. Similar to display retainer **14** in FIG. **6**, retainer **114** releasably secures electronic displays to door frame **114**. Display retainer **114** has a first end **119** coupled to frame rail **113**, and a coupling end **121** extending over electronic display edge **20** to releasably secure electronic display **18** to frame **112**. In some examples, first end **119** is a flange attached to a front surface of frame rail **113**. Flange **119** can be attached to frame rail **113** with one or more mechanical fasteners **117** passing through a front surface of frame rail **113**. Coupling end **121** extends away from first end **119** and is spaced away from a peripheral frame edge **158** of door frame **112**, and away from front surface **60** of panel assembly **11**. To install a electronic display, a electronic display can be placed over panel **11**, and a flange **119** can be placed over both, electronic display edge **20** and frame rail **113**, to secure the flange **119** to frame rail **113** (e.g. with mechanical fastener **117**). To remove a electronic display **18** from door **10**, display retainers **114** securing the electronic display **18** may be quickly removed by removing mechanical fastener **117** from frame rail **113**.

(66) FIG. **8B** depicts a detail view of the portion of the display case door **10** shown in FIG. **6**, according to another example implementation. FIG. **8B** illustrates a panel assembly **11** according to a different embodiment. Panel **11** includes an offset glass package panel assembly **41**. Panel assembly **11** is illustrated as an offset panel assembly. An offset panel assembly **41** can include one more additional transparent panels (e.g., panels **38** and **40**) bonded to a first panel to form an SGU (e.g., panel assembly **11**). In the example arrangement shown in FIG. **8B**, panes **38** and **40** have a smaller surface area than pane **11**, and are bonded to a back surface of pane **11**. Offset panel assembly **41** is not limited to being installed in the implementation of frame **12** shown in FIG. **8B**. Offset panel assembly **41** can be installed in any of the implementations of frame **12** described herein.

(67) FIG. **8C** is a cross-sectional view of the portion of the display case door shown in FIG. **8A**, according to another implementation. FIG. **8C** depicts a frame rail and panel similar to the example shown in FIG. **8A**, with a retainer **114a** configured to accommodate an electronic display **118** with a narrower thickness. For example, each electronic display **118** shown in FIG. **8C** has a thickness smaller than a thickness of frame **112**. A front surface of electronic display **118** is generally

coplanar with a front surface of frame **112**. Display retainer **114a** is a generally flat strip attached to frame rail **113** with a mechanical fastener. Display retainer **114a** has a coupling end **121** that extends over electronic display edge **20** to releasably secure electronic display **118** to frame **112**. In some embodiments, electronic display **118** may be thinner (e.g., having a front surface below frame **112**), in which display retainer **114** can have a coupling end **121** that extends offset from frame **112** toward glass pane **11** (e.g., as shown in FIG. **12** which is described below).

(68) FIG. **9** is a cross-sectional view of the portion of the display case door shown in FIG. **8A**, without electronic displays mounted to the door. The electronic displays can be quickly removed by removing the retainers **114** (shown in FIGS. **8A**, **8B**, and **8C**). A blocking trim **115** can be installed the same way trim **114** is installed in the door frame as shown FIG. **8A**, by attaching flange **119** to a surface of frame **112**. Flange **119** can be attached using a mechanical fastener **117**. Blocking trim **115** covers power cables **50** and electrical connector **54**. Blocking trim **115** may protect cable **50** and connector **54** from the outside environment and may improve the aesthetic appearance of frame **112**. In some implementations, the frame plug **15** shown in FIG. **7** and blocking trim **115** can be used interchangeably. For example, in the arrangement shown in FIG. **9**, frame plug **15** can be used instead of blocking trim **115**.

(69) FIGS. **10** and **11** depict cross-sectional views of a portion of the display case door shown in FIG. **4**, taken along line **10-10** in FIG. **4**, according to two implementations. FIG. **10** illustrates handles **16'** with a J-shaped cross-section. Handles **16'** are attached to doors **10** adjacent to one another. Each handle **16'** is attached to a side surface of a respective door frame **12**. A handle flange **70** is attached to a side surface of frame **12**. Though display retainers **14** are shown securing electronic displays to doors **10**, door trims can also be used instead of display retainers **14**.

(70) FIG. **11** shows door handles **16a** attached to frame **112** through retainers **114**. Handles **16a** can be attached to the door by using an adhesive or mechanical fasteners that pass through the retainer **114** to the door frame **112**.

(71) FIG. **12** depicts a cross-sectional view of the portion of the display case door shown in FIG. **8A**, according to another implementation. FIG. **12** depicts a frame rail and panel similar to the example shown in FIG. **8A**, with a retainer **214** configured to accommodate an electronic display **118** with thickness narrower than frame **12**. Electronic display **218** can be a thin display such as an OLED display. Electronic display **218** is secured to frame **212** to leave a gap between display **218** and panel assembly **11**. A display retainer **214** in the form of a trim piece can be attached to frame **212** to engage electronic display **218** at coupling end **221** of display retainer **214**. Coupling end extends offset from frame **212** toward panel **11**. Frame **212** includes a peripheral frame edge **258** extending from a side of frame **212** to form a channel in cooperation with coupling end **221**, into which electronic display **218** is received and secured.

(72) FIG. **13** is a schematic top view of two adjacent doors **211** and **210**, in which a first door **211** is opened while a second door **210** remains closed. Doors **210** and **211** represent the doors and configuration shown in FIGS. **10** and **11**. Doors **210** and **211** have a cross-sectional shape with an inset **220** that allows a door to open without touching the door that remains closed. The inset corner **220** is representative of the cross-sectional shape formed when the door frames secure an electronic display using the display retainers described in the present disclosure. Because the display retainers are offset from the edge of the door frame, doors **210** and **211** retain a similar thickness around the door edges to existing doors. This may permit users to replace existing doors in a continuous line up of doors (e.g., in a supermarket aisle) without the need to alter the mounting configuration or spacing between doors to accommodate the thickness of the electronic displays. For example, an arrangement where mounted electronic displays increase the thickness of the doors **210** and **211** at the edges may cause the doors to bind against each other if mounted too closely. Such arrangements may require refrigerated display cases to be retrofitted to increase the space between adjacent doors.

(73) FIG. **14** illustrates an exemplary display case door **310** according to another embodiment.

Display case door **310** includes a panel assembly **311** (shown in FIG. **18**), a door frame **312**, covers **313**, electronic displays **18d**, a transparent cover **328**, and a handle **316**. Similar to the door **10** in FIG. **1**, panel assembly **311** can include a single glass panel, a VIG panel, or a glass-filled panel assembly. Electronic displays **18d** are mounted in a one by two configuration, overlaying a portion of panel assembly **311**. In some examples, multiple electronic displays (e.g., more than two) can be arranged in one column that extends along a height of the panel assembly **311**. As further discussed in detail below with respect to FIG. **15**, electronic displays **18d** are mounted on display case door **310** by a mounting arm (not shown) that is attached to door frame **312**. Covers **313** cover the mounting arm and the surface of panel assembly **311** that is not covered by electronic displays **18d**. Covers **313** are attached to frame **312** and extend to a side edge **315** of electronic displays **18d**. Covers **313** are opaque. Covers **313** can be made of plastic or rubber. In some example, covers **313** cover wiring (e.g., power cables **325** and electronic components **326**, **330**). Covers **313** may serve as protection for consumers from the electrical cables and electronic components contained within the door **310** and described in more reference below.

(74) Transparent cover **328** overlaps electronic displays **18d** and can be coupled to door frame **312** or to the electronic displays **18d**. Transparent cover **328** can have a latch (not shown) along one or more edges to snap over an edge of door frame **312** or electronic displays **18d**. In some examples, transparent cover can be attached to door frame **312** using mechanical fasteners. Transparent covers **328** can be made of hard plastic or glass. Transparent cover **328** can be a bezel cover that protects the electronic displays **18d** from damage. For example, a bezel cover may protect the electronic displays **18d** from a consumer bumping the display door **310** with a shopping cart.

(75) FIG. **15** illustrates the door in FIG. **14** showing components (shown in dashed lines) that are covered by covers **313**, electronic displays **18d**, and frame **312**. Each mounting arm **314** has a mounting interface **323** that is configured to secure a respective electronic display **18d** to the mounting arm. Mounting interface **323** is a pattern of fasteners or fastening points that secure electronic display **18d** to mounting arm **314**. More specifically, mounting interface **323** includes a plurality of apertures **324** with mechanical fasteners extending therethrough to thread into corresponding holes (not shown) in a back surface of electronic display **18d**. The pattern of fasteners can be configured to match a corresponding pattern on the back of electronic display **18d**. Each mounting arm **314** has a first end **321a** attached to a first inner flange (shown in FIG. **18**) of frame **312** and a second end **321b** attached to a second inner flange opposite the first flange so that the mounting arm **314** extends along a width of panel assembly **313**. Each of the first and second end of mounting arm **314** are secured to frame **312** by fasteners **320**. Mounting arms **314** can be made of plastic or metal such as aluminum.

(76) Similar to the display case door in FIG. **1**, door frame **312** can include top and bottom openings (not shown) configured to receive a hinge **29** (e.g., an electrical hinge pin) through which a power cable **325** extends. In some implementations, the bottom hinge (not shown) can be a gravity hinge and top hinge **29** can be an electrical hinge pin through which one or more power cables **325** extend. In some implementations, other electrical cables can extend from hinge **29**, such as a conductor to power an anti-condensation system, or a data cable (e.g., a USB or Ethernet cable). Electrical hinge **29** includes an electrical connector **354** (e.g., 120V electrical plug) to which power cable **325** connects to receive power.

(77) Power cable **325** branches into multiple cables that each connect to a respective electronic display **18d** and a power converter **326**. Power cables **325** provide electrical power to electronic displays **18d** and power converter **326**. Power converter **326** receives alternating current (AC) power from power cable **325** and converts the AC power to direct current (DC) power. For example, power converter **326** converts 110V AC power to 5V DC power. Power converter **326** is electrically connected to one or more media players **330**. Power converter **326** is arranged inside the door **310**. For example, power converter **326** can be positioned between the panel assembly **311** and a cover **313**. Power converter **326** can be positioned between the panel assembly **311** and an

electronic display **18d**. In some examples, power converter **326** is coupled to a mounting arm **314**.
(78) Media player **330** can be any type of digital media player device or streaming media player device. Media player **330** is arranged inside the door **310**. For example, media player **330** can be positioned between the panel assembly **311** and a cover **313**. Media player **330** can be positioned between the panel assembly **311** and an electronic display **18d**. In some examples, media player **330** is coupled to a mounting arm **314**.

(79) In some examples, each electronic display **18d** is communicatively connected to a respective media player **330** that is powered by power converter **326**. USB power hub **332** has multiple USB ports that provide power to electronic devices such as cameras and touchscreen devices (shown in FIG. **20**). In some examples, power converter **326** is also electrically connected to a power bar or universal serial bus (USB) power hub **332**.

(80) FIG. **16** shows an implementation of a display case door **410** with the media player **330** being external to display case door **410**. Media player **330** can be secured to a frame of the refrigerated display case (shown in FIG. **4**) or to any surface near display case door **410**. Similar to the door in FIG. **15**, power converter **326** can provide power to media player **330** and a USB power hub **332**.

(81) FIG. **17** illustrates an exemplary display case door **510** according to another embodiment. Display case door **510** includes smaller electronic displays **18e** that, when mounted to door **510**, do not cover areas of the panel assembly **311** above and below the electronic displays **18e**. To cover the exposed areas of the panel assembly, two covers **313a** are placed each adjacent a respective top and bottom edge of electronic displays **18e**, between covers **313**. Covers **313a** can be attached to door frame **312** the same way covers **313** are attached to door frame **312**.

(82) FIG. **18** is a cross-sectional view taken along line **18-18** of a portion of the display case door **310** shown in FIG. **14**. Similar to the cross-sectional view in FIG. **6**, frame rail **317** is configured to receive a bottom hinge **29** and a top hinge (not shown). Either top hinge or bottom hinge can be electrical hinges from which power cable **325** extends. A portion of power cable **325** is disposed within channel **333** and extends from channel **333** between cover **313** and panel assembly **311**. Power cable **325** is electrically connected to electronic displays **18d**. Power converter **326**, media player **330**, and power hub (not shown) can be attached to either one of the mounting arms **314**, the cover **313**, or to the electronic display **18d**.

(83) Cover **313** includes a first end **364** spaced away from an outer peripheral edge of door frame **312** and away from a front surface of panel assembly **311**. Cover **313** additionally includes a second end **358** configured to clip or snug in to an inner surface **362** of door frame **312**. In some examples, first end **364** is flush against an outer peripheral edge of the electronic display **18d**. This arrangement allows cover **313** to cover the surface of panel assembly **311** that is not covered by electronic displays **18d**. Cover **313** can be a bezel cover made of an opaque plastic or a metal. Cover **313** protects the power cables **325** and electronic components (e.g., power converter **326**, media player **330**, and USB hub **332**). Cover **313** may also provide protection for consumers from the power cables and electronic components contained within the door **310**.

(84) Each electronic display **18d** has an outer housing **337** disposed directly against a surface of the mounting arm **314**. One or more mounting arms **314** can support each electronic display **18d**. Each mounting arm **314** has an end **321a** secured to a respective inner flange **360** of door frame **312**. End **321** can be secured to inner flange **360** by using mechanical fasteners **320** such as bolts. In some implementations, a protective foam or flexible tape **331** can separate mounting arm **314** from panel assembly **311**.

(85) FIG. **19** is a cross-sectional view of the portion of the display case door **310** in FIG. **18**, according to a second exemplary embodiment. Cover **313a** has a flange **360a** secured to a front surface of frame rail **317** and an opposite end **364a** extending away from flange **360a**. Opposite end **364a** is adjacent to a side edge of electronic displays **18d**. Flange **360a** is secured to the front surface of frame rail **317** by one or more mechanical fasteners.

(86) Referring back to FIG. **15**, mounting arms **314** and covers **313** allow electronic displays **18d** to

be quickly installed and removed for maintenance or replacement. To install an electronic display, a user can attach mounting arm **314** to a back surface of an electronic display **18d** and, after positioning the electronic display in a desired position over the door, attach each end of the mounting arm **314** to frame **312**. The user can then attach covers **313** to door frame **312** to cover the exposed surfaces of the panel assembly **311** and the mounting arms **314**.

(87) To replace an electronic display, a user can detach covers **313** from door frame **312** and remove mounting arms **314** from door frame **312**. The user can remove electronic display **18d** from mounting arm **314** and mount a new electronic display to arm **314**. A user installs the new electronic display by attaching each end of the mounting arm **314** to frame **312** and then attaching covers **313** to door frame **312**.

(88) FIG. **20** is an electrical diagram of a display case door assembly according to implementations of the present disclosure. Electrical hinge **29** has an electrical connector **329** (e.g., electrical plug) to which power cable **325** is electrically connected. Power cable **32** provides AC power to a door heater **340**, a first electronic display **18d**, a second electronic display **18e**, and a power converter **326**. Door heater **340** can include an anti-condensation system that has an electro-conductive coating to heat the display case door. Two power cables **325d** and **325e** branch out from power cable **325** to provide power to respective electronic displays **18d** and **18e**. Power converter **326** includes a rectifier (not shown) configured to change AC voltage to DC voltage, and a transformer (not shown) configured to decrease the voltage provided by power cable **325**.

(89) Power converter **326** is electrically connected to power hub **332** (e.g., USB power hub) by a power cable **370**. Power converter **326** provides DC power to power hub **332**. Power hub **332** is electrically connected to and provides power to one or more cameras **342** and to one or more touchscreen devices **344**. The power cables **378** that connect cameras **342** and touchscreen devices **344** to power hub **332** can be USB power cords. Cameras **342** can be attached to the door handle or the door frame **312** to detect a customer in front of or near the display case door. Power hub **332** can also provide power to lights (e.g., LED lights) or other electronic devices included in the display case door.

(90) In addition, power converter **326** is electrically connected to respective media players **330d** and **330e**. A first media player **330d** is affixed to or built into first electronic display **18d** and a second media player **330e** is affixed to or built into second electronic display **18e**. Each media player is in data communication with its respective electronic display. Power converter **326** is connected to each media player by respective power cables **372** and **374** that provide DC power to the media players. Media players **330d** and **330e** can be communicatively connected by a cable **376**.

(91) FIG. **21** is an electrical diagram of a display case door assembly according to a second exemplary embodiment. The electrical diagram in FIG. **21** is similar to the electrical diagram in FIG. **20**, with the main exception being that media players **330d** and **330e** are external to electronic displays **18d** and **18e**. Power converter **326** is electrically connected to external media players **330d** and **330e** by power cables **372** and **374**. The media players can be disposed inside the display case door (e.g., between the panel assembly **311** and the cover **313**) or they can be disposed outside the display case door. Each media player **330d** and **330e** is communicatively connected to respective electronic displays **18d** and **18e** by cables **380** and **382**. The media players are communicatively connected to one another by cable **376**. Power converter **326** is electrically connected to power hub **332** by power cable **370**. Power hub **332** provides DC power to electronic devices such as cameras **342** and touchscreen devices **344**.

(92) FIG. **22** shows a front view of a display case door **610** with a single electronic display **18f**. The display **18f** has a width that covers substantially all the panel assembly **311**. In other words, electronic display **18f** is as wide as the interior width of the frame **312**. Electronic display **18f** has a height that is less than the interior height of the frame **312**, leaving a space **612** between the bottom frame member and the bottom edge of the electronic display **18f** to place electrical components

with door **610**. For example, media player **330**, power hub **332**, and power converter **326** can be positioned below electronic display **18f** in the space between the display **18f** and the lower frame member.

(93) Display case door **610** includes a bezel cover **628**. With reference to FIGS. **22**, **23A**, and **23B**, Bezel cover **628** includes a metal or stiff plastic frame **650**, a transparent cover **652**, and an opaque cover **654**. Transparent cover **652** is mounted to the frame **650**. Transparent cover **652** is positioned over the electronic display **18f**. For example, transparent cover **652** can protect the display from damage. Transparent cover **652** can be made of plastic or glass. The opaque cover **654** is mounted to the frame **650**. Opaque cover **654** is positioned over the space **612** between the electronic display **18f** and the bottom frame member of the door **610**. In some examples, bezel cover **628** includes a crossbar **656** mounted across a width of the frame **650**. Crossbar **656** can be positioned along the bezel cover frame **650** so as to be positioned at or near the bottom edge of the electronic display **18f** when the bezel cover **628** is mounted to door **610**. Crossbar **656** can serve as a mounting structure for an edge (e.g., the lower edge) of the transparent cover **652**, an edge (e.g., the upper edge) of the opaque cover **654**, or both.

(94) FIG. **24** illustrates an exemplary display case door **710** according to another embodiment. Display case door **710** includes an insulated panel assembly **711** (e.g., a transparent panel assembly) similar to the transparent panel assembly of FIGS. **1**, **14**, and **22**. Display case door **710** also includes a door frame **712** configured to receive and secure an electronic display to the door. Similar to the display case door of FIG. **22**, display case door **710** has a single electronic display **18g** overlaying the panel assembly **711**. For example, the electronic display **18g** can be sized to fit the interior width of the frame **712**, covering substantially all the panel assembly **711** along its width. In some implementations, electronic display **18g** substantially overlays the panel assembly **711** to cover a majority of the panel assembly. For example, the electronic display **18g** can cover more than 50% of the front surface area of the panel assembly **711**, such as 80% or more. For example, electronic display **18g** has a height that is less than the interior height of the frame **712**, leaving a space **709** between the bottom of the door **710** and the bottom edge of the electronic display **18g** to place electrical components with door **710**. For example, media player **330**, power hub **332**, and power converter **326** can be positioned below electronic display **18g** in the space between the display **18g** and the lower frame member (shown in FIG. **25**).

(95) The door frame **712**, the electronic display **18g**, and the electrical components are covered by respective covers **702** and **704** such as opaque bezel covers. For example, a first door-sized bezel cover **702** is attached to door frame **712** and extends from an outer edge of the frame to an interior edge of a housing of the electronic displays **18g**, exposing only the display screen **39** of the electronic display. A second bezel cover **704** is attached to a bottom surface of the door frame **712** or of the first bezel cover **702** to cover the electrical components.

(96) Electronic display **18g** is releasably secured or coupled to the door frame **712**. FIGS. **27-33** (which are discussed in detail below) illustrate several exemplary implementations of releasable electronic display mounts including, but not limited to: mechanical fasteners extending from the door frame **712** into a housing of the electronic display (FIG. **32**), mounting brackets attached to the housing of the electronic display (FIG. **33**), and a display retainer **714** (e.g., a mounting bezel) secured to the door frame **712** and having a coupling end that extends over an edge of the electronic display **18g** to releasably secure the electronic display to the door frame (FIGS. **27-31**).

(97) Display case door **710** includes a power cable **325** that extends from an electrical connector **354** (e.g., 120V electrical plug) that is part of an electrical hinge **29**. Power cable **29** is disposed within a channel in a portion of door frame **712**. Power cable **325** is connected to and provides power to the power converter **326** and to the electronic display **18g**.

(98) FIG. **25** is an exploded view of an exemplary implementation of a display case door **710**. Electronic display **18g** has an outer housing **37g** and a display panel **39g** (e.g., a screen) mounted to outer housing **37g**. Electronic display **18g** can be positioned directly against the front surface of the

insulated panel assembly **711**, or a protecting foam or flexible tape can be positioned between the back surface of electronic display **18g** and the front surface of panel assembly **711**. Electronic display **18g** can include, but is not limited to, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a field emission display (FED), a plasma display panel (PDP), or an electroluminescent (EL) display. For example, electronic display **18g** can be a smart television with streaming capabilities for receiving content over a wireless network (e.g., a Wi-Fi network). Electronic display **18g** substantially covers panel assembly **711**. For example, the display covers more than 50% of the panel assembly. Electronic display **18g**, together with the circuitry module **737**, overlay the panel assembly **711** to cover substantially the entire surface area of the panel assembly **711**. The surface area of the panel assembly **711** is the front surface area of the panel assembly that is not covered by the door frame **712**.

(99) Display case door **710** includes a transparent protective panel **728** positioned between a front surface **732** of the electronic display **18g** and a coupling end **715** of a mounting bezel **714** type of display mount. Protective panel **728** is similar to the transparent cover of FIG. **14** in that is a protective cover that overlaps the front surface **732** of electronic display **18g** to protect it, for example, from a consumer bumping the display door **710** with a shopping cart. In some examples, the transparent cover **728** can be attached to housing **37g** or an inner surface of the coupling end **715** using adhesive or mechanical fasteners. Transparent covers **728** can be made of hard plastic or glass.

(100) The mounting bezel **714** of display case door **710** features two parallel, vertical members **740** that have a generally L-shaped cross-section (shown in FIG. **27**) and that are separated by two horizontal members **742** that extend from ends of the vertical members **740**. Horizontal members **742** also have a generally L-shape cross-section. The coupling end **715** of mounting bezel **714** can be flange that extends generally parallel to the panel assembly **711**. Mounting bezel **714** has mounting brackets **708** attached to a side of the mounting bezel. Brackets **708** are configured to secure the mounting bezel **714** to the door frame **712**. For example, each mounting bracket **708** has a first end affixed to mounting bezel **714** and a second end, extending away from the first end, configured to interface with the door frame **712** at a respective mounting site **706** of the door frame **712** to secure the mounting bezel **714** to the door frame **712**. As the mounting bezel **714** is secured to the door frame **712**, the coupling end **715** holds the electronic display **18g**, through the protective panel **728**, secured in place by retaining the display against the panel assembly **711**. Thus, the coupling end **715** of each vertical member **740** and the coupling end **715** of the top horizontal member **742** extend over an edge of the electronic display **18g** to cover three edges (e.g., side edges and top edge) of the housing **37g** of the electronic display. Mounting bezel **714** can optionally include multiple louvers **748** configured to provide cooling air-flow to the display **18g** and electronics.

(101) Display case door **710** includes two covers **702** and **704** that, when secured to the door frame **712**, cover the door frame and the mounting bezel **714**. As discussed in detail above with respect to FIG. **24**, the covers **702** and **704** can be bezel covers made of an opaque material. The first bezel cover **702** is sized to overlay the front surface of the door frame **712** and the coupling end **715** (e.g., a coupling flange) of the mounting bezel **714**. The second bezel cover **704** is a rectangular cover that can be attached to the door frame or to the first bezel cover **702** to cover the circuitry module **737** disposed under the electronic display **18g**. The cover bezels can be attached to the door frame with mechanical fasteners, adhesive, or a similar method. Circuitry module **737** includes the media player, power hub, and power converter referred to in FIG. **24**. Circuitry module **737** overlays a portion of the panel assembly **711**. The circuitry module **737** is releasably coupled to door frame **712**. For example, when installed, module **737** can be removed through an opening **703** of the first cover bezel **702**. Circuitry module **737** includes at least a media player (see FIG. **24**) in electronic communication with electronic display **18g**. The media player is configured to control media content presented on display. Second bezel cover **704** covers the circuitry module **737** and can

include multiple louvers **746** configured to provide cooling air-flow to the media player.

(102) FIG. **26A** is a front view of an exemplary display case door **710a** without mounted electronic displays. Display case door **710a** has a panel assembly **711** and a door frame **712** secured to and extending about a perimeter of panel **711**. Door frame **712** is configured to support an electronic display mounted thereto. For example, referring also to FIG. **26B**, the door frame **712** includes a portion **780** that extends outwards a distance beyond a front surface of the panel assembly **711**, and a plurality of mounting sites **706** arranged along the frame **712**. The mounting sites **706** are configured to interface with a display mounting retainer (e.g., with a bracket of a mounting bezel **714**) such that, when mounted, the display (shown in FIG. **24**) overlays the panel assembly. Mounting sites **706** can include open slot cuts of a front surface of the door frame **712** that allow a user to insert a tool to fasten a mechanical fastener used to attach the mounting bezel to the door frame **712**.

(103) As further described in detail below with respect to FIGS. **27-35**, the distance that the portion **780** of the door frame **712** extends outward beyond the front surface of panel assembly **711** can correspond to a thickness of the electronic display and a thickness of the transparent protective panel **728**, such that, when mounted, a front surface of the protective panel is substantially flush with a front surface of the door frame. In some implementations, the distance that the portion **780** of the door frame **712** extends outward beyond the front surface of panel assembly **711** corresponds to a thickness of the electronic display, such that, when mounted, a front surface of the electronic display is substantially flush with a front surface of the door frame.

(104) FIG. **27** is a cross-sectional view of a portion of the display case door shown in FIG. **24**, taken along line **27-27**. For clarity, the power cable and the bottom part of the door frame **712** are not shown. Frame **712** includes a channel **752** into which panel **711** is secured. Panel **711** can be bonded to frame **712** within channel **752** with an adhesive, such as an epoxy or polyurethane. Mounting bracket **708** is affixed to mounting bezel **714**, e.g., mounting bracket **708** can be welded to mounting bezel **714**. The interface between the mounting bracket and the mounting bezel is shown with a dashed line **792**. Mounting bracket **708** has a U-shape cross-section. In some implementations, mounting bracket **708** is made of metal or a hard plastic. In some implementations, mounting bracket **708** and mounting bezel **714** are one piece, such that the mounting bezel or display retainer **714** has a coupling end **715** that mates with the electronic display (e.g., through the protective panel **728**) and a second end **782** that includes a mounting interface **784** that is secured to an inner surface of door frame **12** by a mechanical fastener **790**. Coupling end **715** can be a flange that extends substantially parallel (e.g., within general engineering tolerances within the industry) to panel assembly **711** to releasably secure protective panel **728** and electronic display **18g** to the door frame **712**. Bezel cover **702** is secured to a front surface of door frame **712** by a mechanical fastener **796**. Bezel cover **702** extends from an outer edge of door frame **712** to the edge of the coupling end **715** to cover the door frame **712**, the mounting bracket **708**, and the mounting bezel **714**. Additionally, the outer housing **37g** of electronic display **18g** can be covered by coupling end **715** such that only the display screen **39g** is visible to a customer standing in front of the display case door. In some cases, mounting bracket **708** can have a generally S-shaped cross-section, with the second end **782** extending in the opposite direction (shown with dashed lines). The door handle (not shown) can be attached to a surface of the bezel cover **702** or to a surface of the door frame.

(105) FIG. **28** is a cross-sectional view of the portion of the display case door shown in FIG. **24**, taken along line **27-27**. FIG. **28** illustrates a second exemplary mounting bracket configuration. Though not shown in FIGS. **28-33**, it would be understood that display **18g** includes a housing and a display screen as shown in FIG. **27**, with the edge of the housing being covered by the coupling end **715**, and/or the bezel cover **702**. As illustrated in FIG. **28**, mounting bracket **708a** can have a generally M-shaped cross-section with one end affixed to mounting bezel **714** and a second end **782a** having a coupling interface **784a** that is secured to a back, inner surface of the door frame **712**

by a mechanical fastener.

(106) FIG. 29 is a cross-sectional view of the portion of the display case door shown in FIG. 24, taken along line 27-27. FIG. 29 illustrates a third exemplary mounting bracket configuration. Mounting bracket 708b has a generally L-shaped cross-section that, together with mounting bezel 714, form an S-shape cross-section with coupling end 715 securing the display to the door frame, and the second end 782b secured to a block 795 such as a cast block disposed inside the frame channel of door frame 712.

(107) FIG. 30 is a cross-sectional view of the portion of the display case door shown in FIG. 24, taken along line 27-27. FIG. 30 illustrates a fourth exemplary mounting bracket configuration. Instead of using a mounting bracket, the mounting bezel 714 can be a generally flat member with a coupling end 715 securing the display 18g to the door frame 712, and the second end 782c secured to a cast block by a mechanical fastener 790.

(108) FIG. 31 is a cross-sectional view of the portion of the display case door shown in FIG. 24, taken along line 27-27. FIG. 31 illustrates a fifth exemplary mounting bracket configuration. The display retainer has a generally S-shaped cross-section with a first end secured to a back surface of the electronic display and a second end 782d coupled to the door frame 712 or a cast block 795 of the door frame 712. The display retainer can be a mounting bracket 708d, a mounting bezel, or a combination of the two.

(109) FIG. 32 is a cross-sectional view of the portion of the display case door shown in FIG. 24, taken along line 27-27. FIG. 32 illustrates a sixth exemplary mounting bracket configuration. Instead of being mounted by a mounting bracket or a mounting bezel, electronic display 18g can be secured to door frame 712 by a fastener 791 such as a long mechanical fastener that extends from a side surface of the door frame 712 to a side edge of the housing of the display 18g.

(110) FIG. 33 is a cross-sectional view of the portion of the display case door shown in FIG. 24, taken along line 27-27. FIG. 33 illustrates a seventh exemplary mounting bracket configuration. Door frame 712a can have a generally L-shaped cross-section with a mounting flange 799 extending outwardly, away from panel assembly 711 a distance slightly larger than a thickness of the electronic display 18g. Such distance can include a distance equivalent to a thickness of the display 18g plus a thickness of a transparent protective panel 728a. Protective panel 728a extends from on flange 799 of the door frame 712a, to the opposite flange of the door frame, across substantially the entire width of the display case door. The display retainer can be a mounting bezel or a bracket 708e having a U-shape cross-section, with a first end secured to a side of the housing of display 18g by a first mechanical fastener 790a, and a second end secured to flange 799 of door frame 712a. Mounting bracket 708e has a flat surface substantially flush with a front surface of the electronic display 18g. The transparent protective panel 728a can be secured (e.g., adhered) to the flat surface of the mounting bracket. Protective panel 728a can have an opaque border to cover bracket 708e. The door handle (not shown) can be adhered to the opaque border of the protective panel.

(111) To install electronic display 18g, a user can first obtain a door having a panel assembly and a door frame configured to support an electronic display mounted thereto. The user can position the electronic display over a first portion of the panel assembly and attach at least one display retainer to the door frame. A retainer with a coupling end that mates with an edge of the electronic display, releasably secures the display to the door frame. The circuitry module overlaying a second portion of the panel assembly is secured to the door frame. The media player of the circuitry module is communicably coupled to the electronic display, and the circuitry module is attached to the door frame.

(112) While a number of examples have been described for illustration purposes, the foregoing description is not intended to limit the scope of the invention, which is defined by the scope of the appended claims. There are and will be other examples and modifications within the scope of the following claims. Furthermore, one of skill in the art would appreciate that features described in

reference to a specific embodiment are not limited to that embodiment and can be interchanged with features of other embodiments.

Claims

1. A display case door, comprising: a door frame comprising at least one side that extends along a depth of the display case door towards a front side of the display case door, the door frame configured to retain an electronic video display coupled to the at least one side of the door frame by a plurality of display retainers; and foam insulation housed in the door frame, the foam insulation comprising a front surface, wherein the electronic video display comprises a display screen and the electronic video display overlays a majority of the foam insulation, and wherein the at least one side of the door frame extends forward of the front surface of the foam insulation and is configured to extend alongside a perimeter edge of the electronic video display, when the electronic video display is mounted to the door frame.
2. The display case door of claim 1, wherein the foam insulation is a component of a panel assembly.
3. The display case door of claim 2, further comprising a circuitry module mounted adjacent to a portion of the perimeter edge of the electronic video display, wherein the door frame comprises two horizontal ends spaced apart by a first distance that defines an interior height of the door frame, and wherein the electronic video display and the circuitry module together extend vertically along a second distance that is substantially equal to the interior height of the door frame.
4. The display case door of claim 1, wherein the display retainers comprise mechanical fasteners coupled to threaded coupling sites on the perimeter edge of the electronic video display.
5. The display case door of claim 1, further comprising: a circuitry module comprising a media player in electronic communication with the electronic video display; a power cable electrically connected to the electronic video display and to the media player, and a cover over a front portion of the display case door, the cover comprising a plurality of louvers configured to permit cooling airflow to pass therethrough.
6. The display case door of claim 5, further comprising a plurality of cameras coupled to the display case door and electrically connected to the circuitry module.
7. The display case door of claim 6, wherein the a bezel overlaps a top edge portion of the electronic video display and comprises one or more apertures to promote airflow, and wherein, from a side view, the display case door comprises an inset portion that is inset inward from a maximum external periphery of the display case door, and the inset portion extends outwards in a direction along a thickness of the display case door.
8. The display case door of claim 5, wherein the electronic video display and the circuitry module together overlay an entire surface area of the foam insulation.
9. The display case door of claim 1 further comprising a bezel coupled to the door frame, the bezel configured to overlay and cover a portion of the door frame and a portion of a front surface of the electronic video display, and wherein the display retainers are configured to couple to the perimeter edge of the electronic video display.
10. The display case door of claim 9, wherein the bezel comprises apertures positioned above a top side of the electronic video display and a flange that extends over a portion of a front surface of the electronic video display.
11. The display case door of claim 9, further comprising: a circuitry module mounted below the electronic video display, wherein the door frame comprises two horizontal sides spaced apart by a first distance that defines an interior height of the door frame, and wherein the electronic video display and the circuitry module together extend vertically along a second distance that is substantially equal to the interior height of the door frame; a cover over a front portion of the display case door comprising the circuitry module, the cover comprising a plurality of louvers

configured to permit cooling airflow to pass therethrough; and a plurality of cameras coupled to the display case door and electrically connected to the circuitry module, wherein the bezel comprises apertures positioned above a top side of the electronic video display and a flange that extends over a portion of a front surface of the electronic video display, and wherein, from a side view, the display case door comprises an inset portion that is inset inward from a maximum external periphery of the display case door, and the inset portion extends outwards in a direction along a thickness of the display case door.

12. A display case door, comprising: a door frame comprising two vertical sides and two horizontal sides, wherein a spacing between the two horizontal sides defines an interior height of the door frame, wherein the door frame is configured receive an electronic video display and a circuitry module configured to control the electronic video display; and an opaque insulating material in the frame that covers a region corresponding with at least the interior height of the door frame, wherein the electronic video display and circuitry module, when mounted to the door frame, are positioned external to the opaque insulating material and the circuitry module is positioned adjacent to a portion of a perimeter edge of the electronic video display, and wherein the electronic video display and the circuitry module are positioned together such that they overlay the opaque insulating panel material substantially along a distance that corresponds with the interior height of the door frame.

13. The display case door of claim 12, further comprising: a display retainer coupled between the perimeter edge of the electronic video display and one of the two vertical sides; and a bezel coupled to the door frame, the bezel configured to overlay and cover, with the bezel attached to the display case door, a portion of the door frame and a portion of a front surface of the electronic video display.

14. The display case door of claim 13, wherein the display retainer comprises a mechanical fastener coupled to threaded coupling sites on the perimeter edge of the electronic video display, and wherein the opaque insulating material comprises foam.

15. The display case door of claim 13, wherein the display retainer comprises is attached at one end to the perimeter edge of the electronic video display and attached at a second end to the door frame.

16. The display case door of claim 13, further comprising: a plurality of cameras coupled to the display case door; and a cover positioned over a front portion of the display case door, the cover comprising a plurality of louvers configured to permit cooling airflow to pass therethrough.

17. The display case door of claim 12, further comprising: a display retainer coupled between a perimeter edge of the electronic video display and one of the two vertical sides; a bezel coupled to the door frame, the bezel configured to overlay and cover, with the bezel attached to the display case door, a portion of the door frame and a front surface of the electronic video display; a plurality of cameras coupled to the display case door an in electrical communication with the circuitry module; and a cover positioned over the circuitry module, the cover comprising a plurality of louvers configured to permit cooling airflow to pass therethrough, wherein the display retainer comprises a mechanical fastener coupled to threaded coupling sites on the perimeter edge of the electronic video display, and wherein, from a side view, the display case door comprises an inset portion that is inset inward from a maximum external periphery of the display case door, and the inset portion extends outwards in a direction along a thickness of the display case door.

18. A display case door, comprising: a door support structure comprising two vertical sides spaced apart from one another with an insulating panel supported therein, the door support structure configured to retain an electronic video display mounted to the door supports structure external to the insulating panel, the electronic video display comprising a display screen and overlaying at least part of the insulating panel; a first bezel cover coupled to the door support structure and sized to overlay and cover a portion of the display case door, exposing the display screen; and a second bezel cover comprising louvers, the second bezel cover coupled at a bottom of the door and sized to overlay a circuitry module mounted adjacent to a portion of a perimeter edge of the electronic video display, wherein the insulating panel comprises a surface that is adjacent to and facing a rear

surface of the electronic video display; wherein at least one vertical side of the door support structure comprises: a portion that extends forward of the surface of the insulating panel and is configured to extend along the perimeter edge of the electronic video display by a distance when the electronic video display is mounted, and a plurality of mounting retainers configured to be coupled to the perimeter edge of the electronic video display and to the portion of the vertical side that extends forward of the surface of the insulating panel.

19. The display case door of claim 18, wherein the electronic video display and the circuitry module together overlay substantially an entire height of the insulating panel.

20. The display case door of claim 18, further comprising a plurality of cameras coupled to the display case door, wherein the first bezel cover comprises apertures positioned above a top side of the electronic video display and a flange that extends over a portion of a front surface of the electronic video display, wherein the insulating panel comprises foam and the surface of the insulating panel is a front surface of the foam, wherein, from a side view, the display case door comprises an inset portion that is inset inward from a maximum external periphery of the display case door, and the inset portion extends outwards in a direction along a thickness of the display case door, wherein the mounting retainers comprise mechanical fasteners coupled to threaded coupling sites on the perimeter edge of the electronic video display, and wherein the door support structure comprises two parallel, vertical sides and two parallel, horizontal sides, the two horizontal sides spaced from each other by a first distance that defines an interior height of the door support structure, wherein the electronic video display and the circuitry module together extend vertically along a second distance that is substantially equal to the interior height of the door support structure.
